Nithish K Gnani

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Education

2013 - 2017 **Bachelor of Technology - Mechanical Engineering**, *National Institute of Technology Karnataka (NITK)*, Surathkal, Karnataka, India

Work Experience

2019 - Present **Research Assistant -> Technical Associate**, *Indian Institute of Science (IISc)*, Bangalore
Department of Electronic Systems Engineering & Center for Networked Intelligence

- Selected as a Technical Associate at Cisco - Center for Networked Intelligence (CNI), IISc. Recommended by a committee at CNI, under the division of EECS and Robert Bosch Center for Cyber-Physical Systems (RBCCPS)
- Leading industry-funded novel research projects which also includes planning, and procuring components while judiciously utilizing the available funds.

2017 - 2018 **Team Leader**, *Hindustan Coca-Cola Beverages Private Limited*, Hyderabad
Maintenance and Total Productive Maintenance (TPM)

- Managed a team of 20+ equipment associates to achieve consistent reliability of utility systems.
- Completed the graduate engineering trainee (GET) program with rotations in sales, road-to-market (shipping), and factory operations.

Projects

Mar' 2021 - Present **Time Sensitive Networking (TSN) and Tactile Cyber-Physical Systems**, *Indian Institute of Science*, Funders: Ministry of Electronics and Information Technology (MeitY), Govt. of India & Cisco CNI

- Objective: Design and implementation of a Tactile Cyber-Physical System (TCPS) for real-time interaction between humans and robots for applications requiring ultra-reliable low latency communication (uRLLC).
- Built an IEEE 802.1 TSN capable ethernet switch by implementing in hardware, a. time synchronization (IEEE 802.1AS), b. Time Aware Shaper (IEEE 802.1Qbv), and c. packet duplication and elimination (IEEE. 802.1CB). Built a TCPS testbed with a haptic device (Geomagic Touch) on one end and a robotic arm (UR3 - Universal Robots) on the other end for real time teleoperation.
- *Novelty 1 - P4TAS:* a P4-based system implementation for offloading Time-Aware Shaper onto a programmable SmartNIC. Achieved a latency bound of 20 µs between two end hosts connected through two switches (SmartNICs with P4TAS)
- *Novelty 2:* Developed packet de-duplication algorithms for SmartNICs to efficiently eliminate duplicates for enhancing the reliability of Scheduled Traffic in Time-Sensitive Networks. For duplicating over two links with 10% packet losses in each, achieved perfect de-duplication for a single 2.5 Gbps stream. For 12 simultaneous streams of total 1 Gbps, obtained de-duplication efficiency of 99.88% with 99.83% packet delivery.
- *Novelty 3 - EdgeP4:* Developed and implemented two edge intelligence algorithms for teleoperation, *pose correction* and *tremor suppression* on P4-programmable network edge switches to show that ports can be intelligent in reducing control loop latency (<100 µs for *pose correction* task) and network load (99% reduction). Further, multiple algorithms can be hosted on the same edge switch which can transparently switch between the algorithms depending on the tasks by leveraging P4's match-actions.

Mar' 2020 - June 2022 **AMMAZING**, *Indian Institute of Science*
- An end to end 5G mmWave system for infotainment.
- *Novelty:* Data & control plane of a video stream split over 5G and regular Wi-Fi.
- Shortlisted in top 10 teams in 5G Hackathon 2020-22 by Department of Telecommunication (DoT), Government of India.

July 2020 - June 2021 **Wi-Fi Modelling**, *Indian Institute of Science*, Funder: Boeing (ANRC)
- Modelling & simulation of IEEE 802.11n & 802.11ac using MATLAB.
- Performance Metrics: throughput, packet error rate & range.
- Implementing Rate Control Algorithms in MATLAB.

Feb' 2020 - Jan 2021 **Video Casting in dense environment**, *Indian Institute of Science*, Funder: Boeing
- Considering that 2.4 & 5 GHz Wi-Fi is congested in an aircraft cabin in which 400+ passengers use devices to cast media onto seat back displays in close proximity, the possibility of using IEEE 802.11ad, 60GHz Wi-Fi (Wi-Gig) & 802.11ax (Wi-Fi 6E) was explored.
- WLAN Toolbox in MATLAB was used to simulate the physical behavior of Wi-Fi transmissions in a dense environment.
- *Novelty:* Developed an algorithm that dynamically allocates different channels to 400 transmitters such that the packet error ratio is zero (PER = 0).

June 2019 - Feb' 2020 **Airplane IoT data analytics and management**, *Indian Institute of Science*, Funder: Boeing
- Developed machine learning algorithms to classify anomalies and take corrective actions using real-time data generated from sensors from a MATLAB/Simulink model of an aircraft environmental control system.
- *Novelty:* Developed an algorithm that does linear interpolation between non-consecutive data points from time series data to adaptively store data based on a cost function which balances storage space savings and error in reconstruction of data. Achieved 96% savings in storage space.

Mar' 2019 - Jan' 2020 **Acoustics Based Localization**, *Indian Institute of Science*, Under Dr. Chandramani Singh & Dr. T V Prabhakar, Funder: Boeing
- Developed an application to locate multiple wireless edge devices with embedded microphones by using audio signals from speakers.
- Simulated a few localization algorithms using MATLAB. Implemented KNN fingerprinting based localization of the receivers and achieved an accuracy of 98%.
- *Novelty:* Unlike the usual source localization, here, the receivers (mics) are localized when the locations of speakers are known.

Nov' 2017 - Apr' 2018 **Total Productive Maintenance (TPM)**, *Hindustan Coca-Cola Beverages Pvt. Ltd.*
- Implemented TPM steps 4 and 5 in the bottling plant which led to the JIPM TPM Consistency Award. This included the following work:
- Conducting TPM audits, collecting data, and performing root cause analysis (RCA) for equipment failures
- Training and coaching operators using one-point lesson (OPL) drives
- Planning the cleaning, lubrication, inspection and tightening (CLIT) checklist and schedule

Jul' 2016 - Apr' 2017 **Attitude Control of a CubeSat Using Reaction Wheels**, *National Institute of Technology, Karnataka*, Bachelor's thesis project
- Designed a suspended body and achieved precise control of its yaw using reaction wheels using a manually tuned PID controller.
- Modeled it in Simulink and implemented Model Reference Adaptive Control (MRAC) using the MIT rule and compared it with PID control.

Skills

Programming	P4, Python, MATLAB, C, Embedded C, ROS
Hardware	Netronome Agilio Smart NIC, Nordic nRF52 series, Arduino, NetFPGA-SUME
Networking	TCP/IP, Ethernet, TSN, PTP, WLAN
Others	Autodesk Fusion 360, 3D printing, Adobe Premiere Pro, Implementing papers
Languages	English (IELTS score 8.0), Hindi, Tamil, Kannada, Sourashtra

Publications

In-progress

1. **Nithish Krishnabharathi Gnani**, Joydeep Pal, Deepak Choudhary, Himanshu Verma, Soumya Kanta Rana, Kaushal Mhapsekar, T. V. Prabhakar, Chandramani Singh. EdgeP4: A P4-Programmable Edge Intelligent Ethernet Switch for Tactile Cyber-Physical Systems, (submitted to HotNets 2023)
2. Joydeep Pal, Deepak Choudhary, **Nithish Krishnabharathi Gnani**, Chandramani Singh, T. V. Prabhakar. P4TAS: A P4-based Time-Aware Shaper on SmartNICs, (submitted to 2023 IEEE Global Communications Conference (GLOBECOM))
3. Kaumudi Singh, **Nithish Krishnabharathi Gnani**, Pratyush Shukla, Sachin S M, T V Prabhakar, Joy Kuri, Judicious data management for sustaining an energy harvesting sensor node - extension (manuscript in preparation)

Published

1. Girish Vaidya, T.V.Prabhakar, **Nithish Gnani**, Ryan Shah, Shishir Nagaraja. A novel approach for identification of sensor devices through Acoustic PUF, *Digital Threats: Research and Practice*. https://doi.org/10.1145/3488306
2. Kaumudi Singh, Pratyush Shukla, Sachin S. M., **Nithish K. Gnani**, Prabhakar T. V., Joy Kuri. Judicious data management for sustaining an energy harvesting sensor node, *Concurrency and Computation: Practice and Experience*. https://doi.org/10.1002/cpe.5997

Certifications

1. Python Specialization – University of Michigan.
   https://coursera.org/verify/specialization/9LLM9T3Q55B8
2. Deep Learning – DeepLearning.AI; Ongoing – 4/5 courses complete:
   https://coursera.org/verify/8PG2CRVLTEDH,
   45UFP3FNDV5G, BTQHB5N5CM7V,7899CQP7J2KS

Volunteering

Taught middle and high school students at Yuwa School in Hutup Village in Jharkhand in February 2019. Coached students in tackling interviews. Prepared the team to attend the Laureus Awards 2019, where Yuwa won in "Sports for Good" category.

Interests

Swimming, PC gaming, 3D printing